Temperature, enzymes and metabolism in *E. coli*

What this model adds to what is published

Living Systems Institute, University of Exeter

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The question

- Metabolism is temperature-dependent; **thermal performance curves are the phenotype**
- An **etcGEM** predicts a whole TPC from *per-enzyme* thermodynamics — no curve fitting to the shape
- Three reference points, and every claim here says which one it is relative to:
  - **Li et al. 2021**[1] — the only published etcGEM (yeast)
  - **Pettersen & Almaas 2023**[2] — re-ran it and found the inference unstable
  - **Madkaikar 2023**[3] — the MRes baseline this group inherited
- **What is new here: the data that makes the inference identifiable, and the audits that say what a fitted etcGEM does and does not know**

What an etcGEM is

- A genome-scale metabolic model, plus:
  - **an enzyme budget** — flux costs protein
  - **per-enzyme** $k_{cat}(T)$ — turnover peaks at each enzyme's T_{opt}
  - **per-enzyme unfolding** — the folded fraction collapses at each T_m
- The organism's TPC is then **emergent**, not fitted

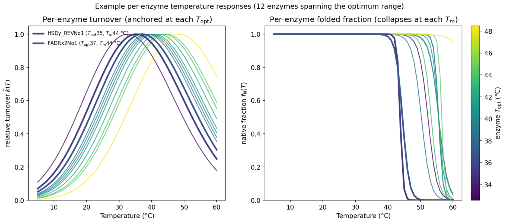

Figure 1: Per-enzyme turnover (left) and folded fraction (right) for ~12 *E. coli* enzymes. `reports/ecoli_tpc`

The state of the art: Li et al. 2021

- Yeast `ecYeast7`, **764 enzymes, 2 292 parameters** (T_m, T_{opt}, $\Delta C_p^{\ddagger}$ each)
- SMC-ABC; T_m **measured** (266 of 764), T_{opt} **predicted** by a sequence model, carried as a wide prior
- Calibrated against **growth and chemostat fluxes**
- **No gas exchange.** No O_2, no acetate
- Result: ERG1 as the flux-controlling enzyme at 40 °C, validated in the wet lab[1]

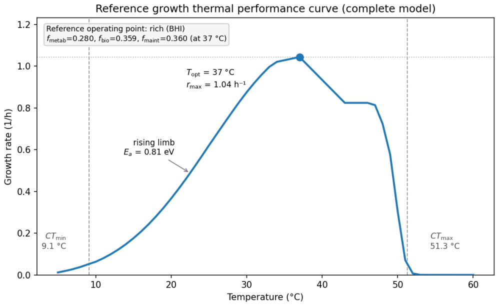

Figure 2: The class of object: a model TPC with T_{opt}, r_{max}, CT_{min}/CT_{max} and E_a marked — here *our E. coli* model at its reference point. `reports/ecoli_tpc`

The problem the field found: Pettersen & Almaas 2023

- *"the Bayesian calculation method … is **unstable and unable to estimate the posterior distribution**"* — it assumes unimodality; the problem is multimodal
- Of six signature reactions, the one that varies most between **equally-fit** particles is **ferrocytochrome-c:oxygen oxidoreductase** — *"for some particles used extensively, in other cases not at all, still giving rise to approximately the same growth rates"*
- Their diagnosis and their ask:
  - *"the external measurements [are] insufficient to account for the inner workings of the … cell"*
  - *"measurements of proteomics and **fluxomics** will help narrow down the solution space"*[2]

The under-determined reaction is the O_2-consuming step of the respiratory chain, and the data they ask for is a measured flux. Hold that thought.

Where we started: the MRes baseline

- `iJO1366`, enzyme-constrained with GECKO 3 in MATLAB; **1 366 enzymes**
- **4 101 thermal parameters** — more per-enzyme freedom than Li et al.[1] — with T_m measured for 841 and T_{opt} from a database for 894
- Same thermal physics as now: two-state unfolding, transition-state k_{cat} with $\Delta C_p^{\ddagger}$, temperature-dependent maintenance
- **Not Bayesian**: repeated FBA scored by R^2 against **one** growth curve, the other 25 held out
- $R^2 = 0.94$; growth TPCs only[3]

Two things to be clear about: the GEM was later **replaced** (`iJO1366` $\rightarrow$ `eciML1515`), and the baseline already had per-enzyme parameters — it had no way to *constrain* them.

What the model is now

- One shared core in Python; **seven organisms** on it (*E. coli* best-constrained)
- **16 global levers**, deliberately — an identifiable set in place of 4 101 unconstrainable ones
- Guarantee against regression: **79/79** structural gates and **60/60** reproduction checks re-run on every change
- The figure is an **a-priori** prediction: nothing fitted to this curve

Emergent model vs Van Derlinden (K-12 MG1655, BHI; raw absolute rate, nothing fit)

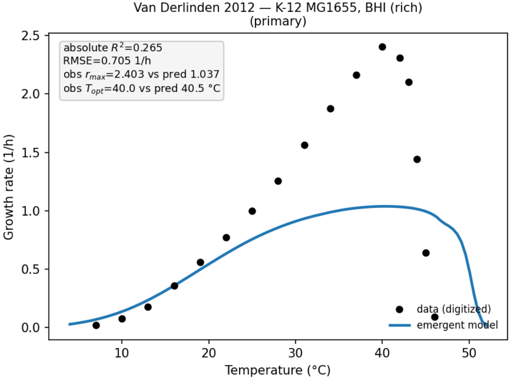

Figure 3: Model against the exact-strain curve (K-12 MG1655, BHI, 7–46 °C; Van Derlinden et al.[4]), predicted with nothing fitted to it. Shape tracks; the peak sits ~2.3-fold low.

Development 1: proteome sectors and the growth law

- Enzyme budget split into **metabolic / biosynthetic / maintenance** sectors[5,6]
- Sector fractions are **measured against temperature** — the chaperone sector ramps above 37 °C
- New relative to Li et al.[1] and to the MRes[3]: neither has an allocation layer
- Consequence, and a caution: with sectors **temperature-independent** the top of the TPC goes flat and T_{opt} is the argmax of a tie

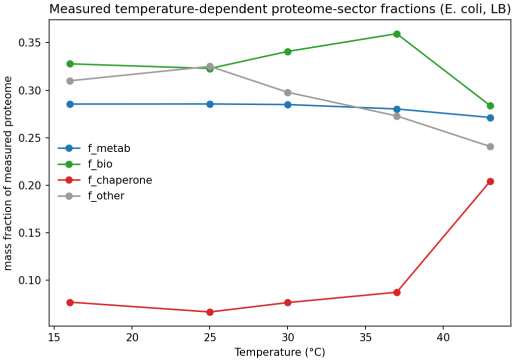

Figure 4: Measured *E. coli* proteome-sector mass fractions across LB temperature; chaperones in red. `reports/ecoli_tpc`

Which lever owns which feature

- r_{max} is owned by **catalysis** (σ, k_{cat} scale)
- CT_{max} and T_{opt} by **stability** (dT_m, tm scale)
- E_a and niche width by the **spread of enzyme optima**
- This is why a TPC descriptor is not a parameter: **each one is a different question of the model**
- Deterministic elasticities at the operating point — **no intervals on this slide**

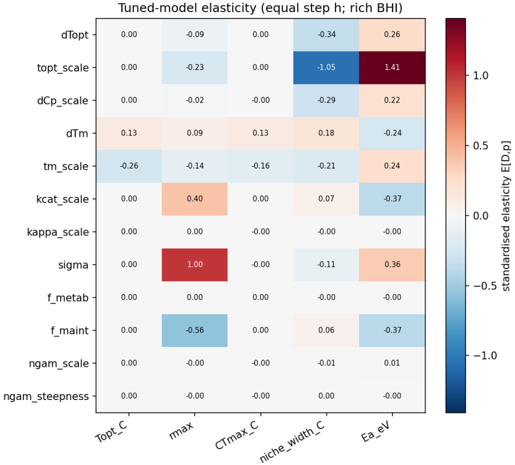

Figure 5: Elasticity of every curve descriptor to every lever, tuned point, standardised step. reports/ecoli_tpc

Development 2: overflow metabolism, emergent

- Acetate overflow is **not imposed**: it falls out of a **total-carbon-uptake cap** plus the enzyme budget
- Ported from Parsa Amirmoeini's configurations A–F and **gated**: his numbers reproduce here (60/60, and 10/10 on the respirometry fits)
- Also ported: an **ETC membrane-area** constraint — configurations E and F collapse to one mechanism
- **This slide is mechanism.** The fit quality behind it is a point estimate from an unconverged chain; no interval is quoted

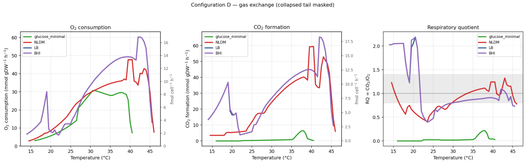

Figure 6: Configuration D: predicted O_2 uptake and acetate against temperature, over the measurements. **Mechanism, not interval.** `reports/ecoli_gasflux`

The data nobody else has

- Parsa's respirometry: **O_2 flux and growth per cell, 15–50 °C, three media** (R2A, LB, M9), ∼60 replicate runs each
- This is exactly the **fluxomics on the O_2-consuming reaction** that Pettersen & Almaas[2] name as missing
- And it is a **measurement**, not a model output
- Note the shape: **respiration peaks 3–5 °C above growth** (43–45 °C vs ∼40 °C) and is still rising where growth has begun to fail

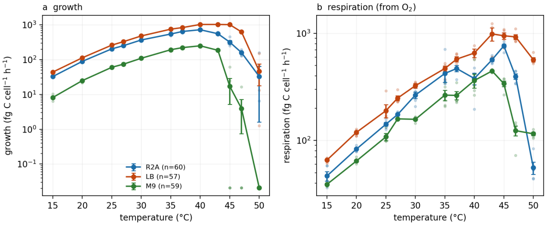

Figure 7: Measured per-cell growth (a) and respiration (b) against temperature, three media. Points are replicates, bars are s.e.m. Drawn here from `strains/eciML1515/respirometry/derived_*_cur`

…and what it costs the cell

- **Carbon-use efficiency** — the fraction of assimilated carbon retained as biomass
- Flat at ∼**0.6** (R2A, LB) and ∼**0.4** (M9) from 25 to 40 °C
- Then a **cliff**: 0.29 / 0.52 / 0.05 at 45 °C
- CUE is listed in the *E. coli* paper as **future work**. It is measured, in three media, and not yet folded in
- Medium sets the *level*; temperature sets the *cliff*

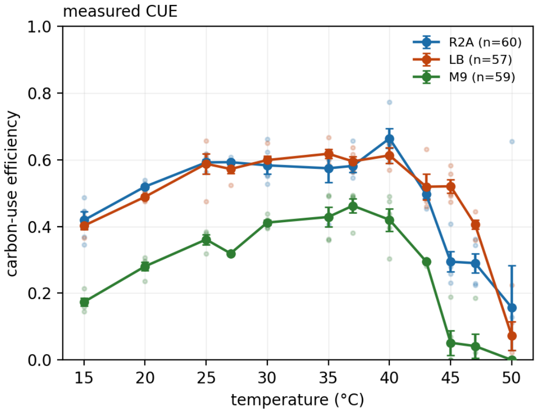

Figure 8: Measured carbon-use efficiency against temperature, three media, same tables and same script.

Calibrating it: the surface is cliffed, not under-sampled — and the fixes

- 22 line scans, 902 fresh model builds: **12 of 22 lines** move >20 % of their range on a small step
- **Not the solver**: unchanged at 10 tolerances and on dual simplex
- Each cliff is the **respiration term at one cold temperature**, where O_2 at the growth optimum jumps LP vertex
- Three fixes, each for a named mechanism: a **tie-break** on the degenerate face; a **variance floor** the model can honour; a **continuous support** clamp at the growth mask
- Cliffs cut $\sim 10\times$; surface still **not smooth**. Reported as such

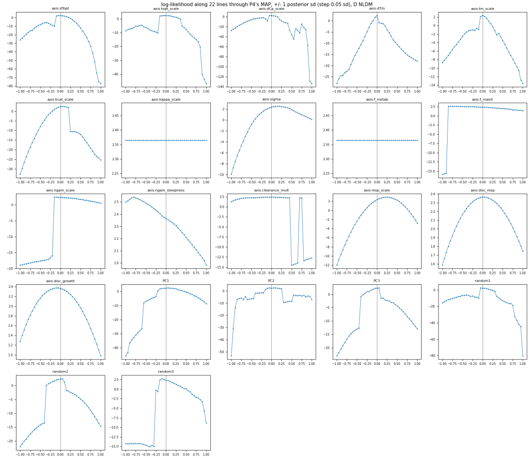

Figure 9: Likelihood scanned along parameter lines: parabolas separated by cliffs, and two axes exactly flat. `reports/P9_surface`

One live basin — and two retractions

- **P12** mapped the basins: of twelve converged endpoints, **one live basin** and one dead one. Not the many-modes picture Pettersen & Almaas[2] found in yeast — but also not a posterior
- **Retraction 1 (ours).** P12 concluded the meltome-honouring region *“contains no growing model”*. **It does**: with the melting-temperature shift and spread both pinned to the measured meltome, the fit converges and the organism grows at 1.66 h ^{1} — 7.6 log-likelihood units worse, and pressed against two prior ceilings
- **Retraction 2 (ours).** P12’s ceiling test was qualified by P13: five of twelve endpoints park a temperature within 1 % of the growth mask, and the mask is an **attractor**, so part of the margin was an artefact
- **Where the posterior stands:** two nested runs are **in flight**. The criterion is fixed in advance — chain length $>$ 40 autocorrelation times and $n_{eff} \geq 200$. **Nothing is quoted from them here**

Auditing the reference implementation

- **Y1** — we built a coupling-ion audit and ran it on Li et al.'s deposited model[1]. **The defect is absent**: their respiratory chain supplies **93 %** of ATP synthase's protons, all twelve uncosted proton-movers run the dissipative way, and no free path exists at any length
- **Y2** — their published *posterior*, 100 models, their own code: switching the binding constraint widens the top of the TPC from **1.5 to 7.3 °C in 93 %** of them
- But CT_{max} **also** moves (4.6 °C) — in *our* model it did not. **The asymmetry is not the clean thing we thought**

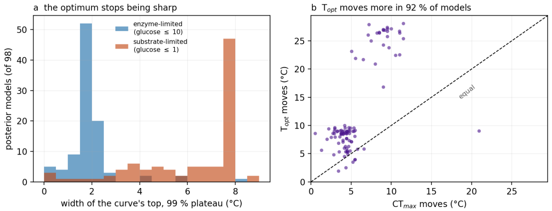

Figure 10: Li et al.'s own 100 posterior models under a change of binding constraint. Drawn from `reports/Y2_regime_posterior` committed tables.

The awkward one: our model needs a −4 K stability shift

- Every fit of the *E. coli* model needs melting temperatures **3–4.5 K below a measured meltome**
- Screened: is that biology (*in vitro* T_m is not *in vivo* inactivation) or **our parameterisation** (16 global levers where Li et al.[1] had 2 292)?
- **The shift is not uniquely required** — an equally good, living fit exists at zero shift, for 0.08 log-likelihood units
- **But catalysis is not what pays for it.** The compensator is the *spread* of melting temperatures. Both solutions put the least-stable few per cent of enzymes **4–6 °C below measurement**
- So: the **number** is a parameterisation artefact; the **contradiction with the meltome's low tail** is not. Per-enzyme $k_{cat}(T)$ would not fix it — and the predictor we would have used has no usable optimum for 97 % of these enzymes

Settled · in flight · needs measurement

- **Settled.** The shared core and the seven-organism port, gated. The mechanisms: sectors, overflow from a carbon cap, ETC area. The audits: the coupling-ion class, and the reference implementation passing it
- **Settled, and uncomfortable.** No interval from the older *E. coli* fits is available. The likelihood surface is structurally rough. The meltome's low tail is contradicted by every fit
- **In flight.** The posterior, on a pre-registered convergence criterion
- **Needs measurement — not more modelling.** Proteome allocation against temperature; maintenance against temperature from a low-dilution chemostat; and *in vivo* stability for the enzymes that actually set the thermal limit
- **Never yet tested.** Nothing in this framework has been asked to predict data it was not fitted to. Cooper et al.'s evolved *E. coli* lines[7] are the obvious holdout

Three questions for the group

1. **Is a few-kelvin gap between an *in vitro* meltome and *in vivo* functional inactivation expected?** If it is, the model is asking the meltome the wrong question and we should say so. If it is not, the model is wrong somewhere we have not looked
2. **Respiration peaks 3–5 °C above growth and CUE falls off a cliff at 43–45 °C.** Is that a general property of thermal performance in bacteria, or is it a feature of these three media?
3. **What would you accept as a real test?** Our candidate is predicting how a TPC *shifts* after 20 000 generations of thermal selection[7]. Is there a better one?

References I

1. Li, G. *et al.* Bayesian genome scale modelling identifies thermal determinants of yeast metabolism. *Nature Communications* **12**, 190 (2021).
2. Pettersen, J. P. & Almaas, E. Parameter inference for enzyme and temperature constrained genome-scale models. *Scientific Reports* **13**, 6079 (2023).
3. Madkaikar, A. Predicting the thermal niche of a ubiquitous bacterium using whole genome sequence. (Imperial College London, 2023).
4. Van Derlinden, E. & Van Impe, J. F. Modeling growth rates as a function of temperature: Model performance evaluation with focus on the suboptimal temperature range. *International Journal of Food Microbiology* **158**, 73–78 (2012).
5. Basan, M. *et al.* Overflow metabolism in Escherichia coli results from efficient proteome allocation. *Nature* **528**, 99–104 (2015).

References II

6. Scott, M., Gunderson, C. W., Mateescu, E. M., Zhang, Z. & Hwa, T. Interdependence of cell growth and gene expression: Origins and consequences. *Science* **330**, 1099–1102 (2010).

7. Cooper, V. S., Bennett, A. F. & Lenski, R. E. Evolution of thermal dependence of growth rate of Escherichia coli populations during 20,000 generations in a constant environment. *Evolution* **55**, 889–896 (2001).